PERSONAL LOAN

📘 Business Requirements Specification (BRS)

Personal Loan Management System – Bank Based

Version 1.0

1. Document Control

Document Title: Business Requirements Specification – Personal Loan Management System

Version: 1.0

Date: *[Auto-populated]*

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Approved By: Project Sponsor

1.1 Revision History

Version	Date	Description	Author
1.0	10/12/2025	Initial Release	BA

2. Purpose / Objective

The purpose of the **Personal Loan Management System (PLMS)** is to digitalize and streamline the **end-to-end personal loan lifecycle** for a bank.

The system will enable:

- Faster loan processing
- Digital documentation
- Automated eligibility checks
- Transparent tracking
- Regulatory-compliant workflows
- Secure document handling
- Improved customer experience

This BRS defines the **business needs** clearly — independent of any technology or design — and serves as the foundation for SRS, architecture, and development.

3. Business Background / Context

Banks currently handle personal loan processing through:

- Manual document verification
- Paper-based workflows
- Multiple branch-level approvals
- Telephone-based customer communication

- Longer decision cycles

These create challenges such as:

- Delayed loan approval
- High operational cost
- Low transparency for customers
- Difficulty in tracking loan status
- Inefficiency in document management

The new **Personal Loan Management System** aims to:

- Automate workflows
- Reduce manual dependencies
- Improve customer experience
- Ensure compliance with RBI guidelines
- Enable faster disbursement

4. Scope

4.1 In-Scope

The system will include:

Customer Module

- Customer onboarding
- Loan application
- Document upload
- Status tracking
- EMI schedule & payments
- Support requests

Back-Office Modules

- Loan Officer Portal
- Underwriter Portal
- Verification Team Portal
- Disbursement Portal
- Collection Team Portal
- Admin + Super Admin Portal

System Integrations

- PAN/Aadhaar Verification API
- CIBIL Score API
- SMS/Email Notifications
- Payment Gateway

4.2 Out-of-Scope

- Credit card issuance
- Other loan types (home/auto/business)
- Branch-based offline workflow
- Mobile App (Phase 2)
- Cash-based EMI collection

5. Stakeholder List

Stakeholder	Description
Customer	Applies for loan
Loan Officer	Reviews applications

Stakeholder	Description
Verification Team	Validates KYC & documents
Underwriter	Performs risk assessment
Disbursement Officer	Disburses approved loans
Collection Team	Manages overdue EMIs
Admin	Manages users & loan products
Super Admin	System-level configurations

6. User Roles & Access

Role	Access Permissions
Customer	Apply loan, upload docs, track status, pay EMI
Loan Officer	View, approve/reject, request documents
Verification Team	KYC verification, document validation
Underwriter	Eligibility analysis, risk evaluation
Disbursement Officer	Final disbursement approval
Collection Team	EMI tracking, reminders
Admin	Manage users, products, interest rate
Super Admin	Full access to system configuration

7. Requirement Gathering Summary

Requirements were collected through:

- Stakeholder meetings
- Banking process study
- RBI personal loan compliance review
- Existing loan policy documents

- Verification team interviews
- Customer journey analysis

Key findings:

- Customers demand speed & transparency
- Bank requires strict compliance and security
- Automation is essential to reduce turnaround time (TAT)
- Audit logs required for all loan decisions

8. Business Requirements

8.1 Customer Requirements

BR-C1: Customer Registration & Login

- Customer must be able to create an account using mobile number, email, and OTP verification.
- Login via username/password with multi-factor authentication.

BR-C2: Loan Application Submission

- Customer must be able to apply for a personal loan by providing:
  - Personal details
  - Employment details
  - Income details
  - Bank details
  - Loan amount & tenure

BR-C3: Document Upload

- Customer must upload mandatory documents (PAN, Aadhaar, salary slips, bank statement, photo).
- System should allow re-upload on request from the back-office team.

BR-C4: Track Loan Status

- Customer must see real-time status:
  - Submitted → Under Verification → Underwriting → Approved/Rejected → Disbursed → Active

BR-C5: View EMI Schedule & Loan Ledger

- Customer must view a complete EMI schedule.
- Customer must view month-wise loan ledger (interest, principal, outstanding).
- Customer must download ledger & EMI schedule as PDF.

BR-C6: EMI Payment

- Customer must be able to pay EMI through:
  - UPI
  - Net Banking
  - Debit Card
  - Auto-Debit/ACH/NACH

BR-C7: Support Requests

- Customer can raise queries, track support tickets, and communicate with support team.

8.2 Loan Officer Requirements

BR-LO1: View Loan Applications

- Loan officer must view all applications with status and risk flags.

BR-LO2: Document Verification

- Validate customer documents.
- Request resubmission if documents are unclear or invalid.

BR-LO3: Soft Approval/Rejection

- Loan officer may:
  - Approve for underwriting
  - Reject with reason
  - Put on hold

BR-LO4: Update Application Notes

- Officer must be able to add internal comments visible to underwriters/admin.

8.3 Verification Team Requirements

BR-V1: KYC Verification

- Validate identity through Aadhaar/PAN verification APIs.

BR-V2: Employment Verification

- Verify employer details, salary slips, and bank statements.

BR-V3: Address Verification

- Perform online/address check depending on policy.

BR-V4: Fraud/Risk Checks

- Validate blacklists, duplicate applicants, and risky profiles.

8.4 Underwriter Requirements

BR-UW1: Risk Assessment

- Evaluate:
  - CIBIL score

    - Income-to-EMI ratio
    - Banking behavior
    - Repayment capacity

BR-UW2: Final Approval/Rejection

- Underwriter gives final decision.
- System must track decision timestamps and approver details.

BR-UW3: Modify Loan Terms

- Underwriter may modify:
    - Interest rate
    - Tenure
    - Loan amount
    - Repayment type

8.5 Disbursement Team Requirements

BR-D1: Disbursement Processing

- After approval, system generates:
    - Sanction letter
    - Loan agreement
    - Repayment schedule

BR-D2: Bank Account Validation

- Validate customer's bank account via penny-drop or similar API.

BR-D3: Transfer Funds

- Initiate disbursement via NEFT/IMPS or internal core banking.

8.6 Ledger & Repayment Requirements

BR-L1: Ledger Generation

- System must automatically generate a loan ledger on disbursement.
- Ledger must follow amortization rules (interest first, then principal).

BR-L2: Ledger Updates

- Ledger updates on:
  - EMI payment
  - Part-payment
  - Foreclosure
  - Penalty posting
  - Interest rate changes (floating loans)

BR-L3: EMI Posting

- EMI posting:
  - First adjusts interest
  - Then principal
  - Then penalties (if any)
- Outstanding balance must update accurately.

BR-L4: Overdue Management

- System must identify overdue EMIs.
- Calculate penalties/late fees as per bank policy.

BR-L5: Prepayment & Foreclosure

- System must:
  - Allow part-payment
  - Allow early closure

    - Recalculate ledger & outstanding
    - Generate closure certificate

8.7 Collection Team Requirements

BR-CO1: Overdue EMI Tracking

- Collection team must view overdue customers with aging buckets:
    - 0–30 days
    - 31–60 days
    - 61–90 days
    - 90+ NPA indicators

BR-CO2: Communication Logs

- Team must log:
    - Calls
    - SMS
    - Notices

BR-CO3: Payment Arrangements

- Team can update:
    - Promise-to-pay (PTP)
    - Settlement offers (with admin approval)

8.8 Admin & Super Admin Requirements

BR-A1: User Management

- Create/modify/delete users and assign roles.

BR-A2: Loan Product Management

- Define:
    - Interest rates
    - Processing fees
    - Tenure options
    - Prepayment rules

BR-A3: System Configuration

- Configure:
    - Email/SMS templates
    - Loan approval limits
    - Risk rules
    - Notification settings

BR-A4: Audit & Compliance

- Track:
    - All activity logs
    - Change history
    - Compliance reports (RBI)

8.9 System Integration Requirements

BR-SI1: CIBIL Integration

- Fetch credit score directly from the CIBIL API.

BR-SI2: Aadhaar/PAN Verification

- Validate identity using government verification APIs.

BR-SI3: Payment Gateway Integration

- Support online EMI payments via 3rd-party gateways.

BR-SI4: SMS/Email Notification System

- Auto-trigger notifications at:
  - Application submission
  - Approval/Rejection
  - EMI reminders
  - Overdue alerts
  - Disbursement updates

8.10 Reporting Requirements

BR-R1: Operational Reports

- Applications report
- Disbursement report
- Overdue report
- NPA report

BR-R2: Ledger Reports

- Month-wise loan ledger
- Customer ledger summary
- Interest earned
- Principal recovered

Non-Functional Requirements (NFR)

Performance

- All APIs should respond < **2 seconds**
- System must support **10,000 concurrent users**

Security

- Data encrypted (AES-256)
- API secured via JWT
- Documents stored securely
- Audit logs for every update

Availability

- 99.9% uptime

Scalability

- Horizontally scalable microservices

Compliance

- RBI guidelines
- KYC/AML norms
- Data retention policies

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9. Business Rules (Expanded)

- Interest rate set by Admin only
- Processing fee auto-calculated
- Late penalty: 2% per month
- EMI auto-debit date = disbursement date + 30 days
- Loan cannot be disbursed without KYC approval

10. Assumptions

- Interest rates may change monthly
- API providers remain active

- Customer has valid mobile number
- Bank maintains KYC records

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11. Constraints

- Must follow RBI KYC norms
- Aadhaar masking required
- Maximum file upload = 5MB
- Java 17 + Spring Boot 4

12. Software & Technical Requirements

Category	Technology
Language	Java 17
Framework	Spring Boot
DB	MySQL
IDE	Spring Tool
VCS	Git / GitHub
Build	Maven
Cloud	AWS
APIs	CIBIL, PAN, Aadhaar

13. Data Requirements (High-Level)

Customer Data

- Name
- DOB

- Address
- Mobile number
- Email

Loan Data

- Loan ID
- Amount
- Tenure
- Interest
- Status

KYC Data

- PAN
- Aadhaar
- Salary documents

14. UI/UX Requirements (High Level)

- Clean customer onboarding screen
- Multi-step loan application form
- Real-time validation
- Dashboard with loan timeline
- EMI schedule page

15. Workflow Diagrams (Described)

1. Customer applies →

2. Document upload →

3. KYC verification →

4. CIBIL check →

5. Loan officer review →

6. Underwriter approval →

7. Sanction letter →

8. Disbursement →

9. EMI generation & payment

16. Integration Requirements

- CIBIL REST API
- Aadhaar/PAN verification API
- Payment Gateway
- SMS & Email Service

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17. Reporting Requirements

- Daily loan disbursal report
- EMI collection report
- Overdue aging report
- Customer KYC report

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18. Acceptance Criteria

- End-to-end loan workflow must function
- EMI schedule must be mathematically correct
- All roles must access only allowed features

- Every loan decision must be auditable

19. Risks & Mitigation

Risk	Mitigation
API downtime	Retry mechanism
Document fraud	Advanced validation
High user load	Auto scaling
Payment failures	Retry + customer notification

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20. Glossary

- **CIBIL:** Credit score in India
- **KYC:** Know Your Customer
- **AML:** Anti-Money Laundering
- **EMI:** Monthly loan installment
- **Underwriting:** Risk evaluation process

21. Appendix

- Sample Salary Slip
- Sample Sanction Letter
- Sample EMI Table

✅ How to Calculate EMI (Equated Monthly Installment)

Banks use a **standard formula** for EMI:

📌 EMI Formula

EMI=P×R×(1+R)N(1+R)N−1EMI = \frac{P \times R \times (1+R)^N}{(1+R)^N - 1}

EMI=(1+R)N−1P×R×(1+R)N

Where:

Symbol	Meaning
P	Principal amount (Loan amount)
R	Monthly interest rate (Annual Rate / 12 / 100)
N	Tenure in months

🔍 Example Calculation

Loan Amount (P) = ₹1,00,000

Annual Interest Rate = 12%

Tenure = 12 months

Step 1 → Convert rate to monthly

R=1212×100=0.01R = \frac{12}{12 \times 100} = 0.01

R=12×10012=0.01

Step 2 → Put in formula

EMI=100000×0.01×(1.01)12(1.01)12−1EMI = \frac{100000 \times 0.01 \times (1.01)^{12}}{(1.01)^{12}-1}

EMI=(1.01)12−1100000×0.01×(1.01)12

EMI=₹8,885EMI = ₹8,885

EMI=₹8,885

So monthly EMI = **₹8,885**